

1. (10%) For virtual memory

- Does it use set associative, fully associative or direct map scheme? Why?
- What is the write strategy? Write back or write through? Why?

2. (20%) Briefly and clearly explain the following terms:

- Translation Look-aside buffer
- Non-blocking cache
- Can non-blocking cache reduce miss rate? Why or why not?

3. (15%) (a) What is the protection mechanism of RAID 4? Explain how it handles small writes.

(b) What is the protection mechanism of RAID 6?

4. (15%) Suppose the length of the address is 32 bits. For a 4-way set associative 512KB cache whose block is 128 Bytes, how many bits are used for index? How many bits are used for tag?

7. (10%) About cache design, which of the following statements are correct?

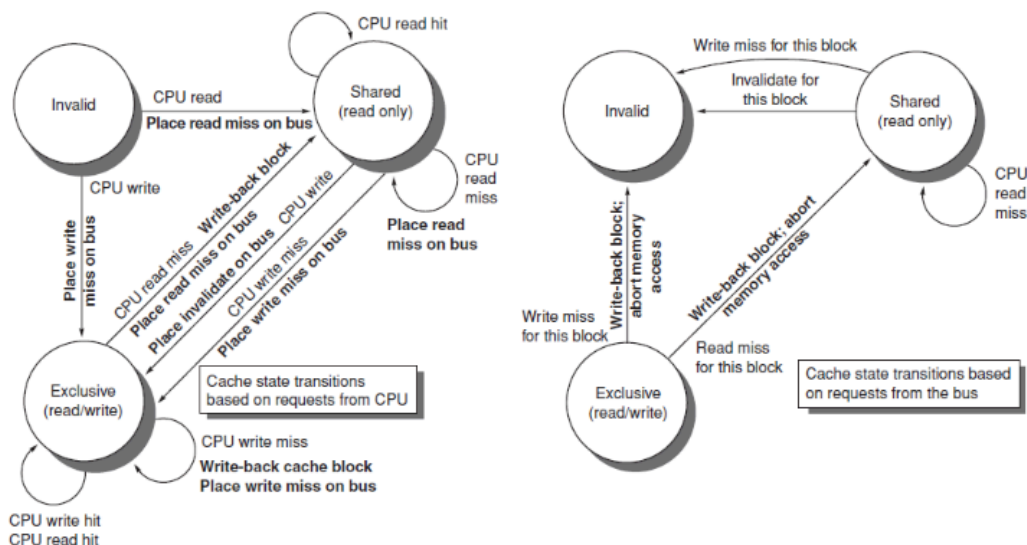
- Lower level of associativity reduces conflict miss.
- Larger block size reduces capacity miss.

6. (10%) Suppose you want to achieve a speedup of 5 with 10 processors. At most what fraction of the original computation can be sequential?

- 5%
- 11%
- 15%
- 17%

5. (10%) The state transition diagrams of a write invalidate, cache coherence protocol for a write-back cache are illustrated below. Suppose two processors P1 and P2 are both in Shared state for a data block X. P2 writes X at time T1. Then, P1 writes X at time T2. Afterwards, P2 reads X at time T3. Which of the following statements are true?

- At time T1, P2 sends a CPU write request and place invalidate on bus and then goes to exclusive state.
- At time T1, P2 sees write miss for this block from the bus and goes to invalid state.
- At time T2, P1 sends a CPU write request and place write miss on bus and then goes to shared state.
- At time T2, P2 sees write miss for this block from the bus, it writes back the block, abort memory access and goes to invalid state.
- At time T3, P1 sees read miss for this block from the bus, it writes back the block, abort memory access and goes to shared state.



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- (b) Non-blocking cache
- (c) Can non-blocking cache reduce miss rate? Why or why not?

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7. (10%) About cache design, which of the following statements are correct?

- (A) Lower level of associativity reduces conflict miss.
- (B) Larger block size reduces capacity miss.

8. (10%) Which of the following statements are NOT true?

- (A) For a cache with write back strategy, read misses might result in writes.
- (B) For a cache with write through strategy, a write buffer is not necessary.
- (C) Write after write (WAW) hazards can be resolved by register renaming.
- (D) Read after write (RAW) hazards can be resolved by register renaming.
- (E) Distributed shared-memory scheme will not result in non-uniform memory access time for multiprocessor machines.